OSPE DAY 1

SPOT NO. 1.

1. Identify the A, B, and C?
   Ans: A → Plica circularis; B → Intestinal villus; C → Brush border
2. How much do each increase the surface area of intestine?
   Ans: A → 5 – 8 times; B → 40 times; C → 600 times.
3. Secretions poured by small intestine contain which type of enzymes?
   Ans: They contain no enzymes. Enzymes are present on brush border epithelium.

SPOT NO. 2.

1. How many types of cells are present in the small intestinal mucosa?
   Ans: Stem cells, Enteroendocrine cells, Paneth cells, Absorptive cells, Mucus cells
2. What is the function of Paneth cell?
   Ans: They have antimicrobial functions regulating the flora of intestine
3. How much fluid is absorbed from the small intestine during the passage of chyme?
   Ans: 7.5 L

SPOT NO. 3.

1. Write down the names of hormone secreted by each cell:
   Ans: I cells → CCK; K cells → Gastric Inhibitory Polypeptide (GIP); L cells → Glucagon-like Peptide 1 (GLP-1) and Peptide YY (PYY); N cells → Neurotensin
2. What are the functions of hormone secreted by Mo cells?
   Ans: Motilin stimulates gastric and small intestine motility, causing undigested food in these regions to move into the large intestine. Cyclically released during fasted state.
3. What are the functions of hormone secreted by I cells?
   Ans: 1. Contraction of Gallbladder; 2. Relaxation of sphincter of Oddi; 3. Inhibits gastric emptying; 4. Stimulates acinar cells of pancreas to release a juice rich in pancreatic digestive enzymes

SPOT NO. 4.

1. What is the importance of CFTR channel protein?
   Ans: Cystic fibrosis transmembrane conductance regulator (CFTR) protein helps in maintaining the integrity of intestinal mucosa by keeping the mucosa hydrated
2. Name a condition where CFTR protein is not working properly, trapping Cl- inside the cell.
   Ans: Cystic fibrosis
3. Which toxin stimulates this protein causing secretory diarrhea
   Ans: Vibrio Cholera toxin

SPOT NO. 5.

1. Which two types of movements occur in the small intestine?
   Ans: Segmentation contractions; Migratory motility complex
2. Which two types of movements occur in the large intestine?
   Ans: Haustration contractions (or Haustral movements); Mass movements
3. What is gastrocolic reflex?
   Ans: The gastrocolic reflex is a physiological reflex that controls the motility of the lower gastrointestinal tract following a meal